Supplementary Document

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1 Supplement Reader Crosswalk

This crosswalk explains how the supplement should be read with the main article. Each row identifies the main article surface, the supplement section that supplies the detailed audit evidence, the primary source artifact, and the stronger claim that remains beyond this study. The table is a navigation aid; it does not add experiments or promote a method.

Main article surface	Supplement support	Primary source artifact	Stronger claim requiring validation
CQR, Mondrian, and CV+ descriptive performance	S1 robustness diagnostics, S1b CQR backend sensitivity, and S2 post-selection diagnostics	`method_selection_robustness_audit; cqr_fixed_vs_model_matched_synthesis`	No study-wide method choice, global best method, or deployment rule.
Venn-Abers bridge negative evidence	S2 bridge validation slice and S6 traceability	`dataset_final_gate_post_selection_validation_bridge_results`	No validated Venn-Abers regression interval claim.
Bounded-support validity	S3 bounded-support endpoint policy	`bounded_support_endpoint_closure_audit`	No endpoint or bounded-support validity claim.
Population-level group inference	S4 group diagnostics	`group inference_population_readiness_audit`	No population-level group inference claim.
Duplicate and leakage integrity	S5 duplicate and integrity caveats	`cross_run_integrity_audit`	No claim strengthening by hiding duplicate or split-integrity caveats.
Knowledge-graph traceability	S6 traceability	`knowledge_graph_quality_summary`	The KG and site support navigation and source tracing.

2 Supplement Reading Protocol

The supplement is intentionally broader than the main article. Each section adds audit detail, but the extra detail must be read through a specific boundary. The protocol below states what a reviewer should check, which evidence metric anchors the check, the allowed use of that evidence, and the overclaim that remains beyond this study.

Section	Reviewer check	Evidence metric	Allowed use	Beyond this study use
S1	Check whether the method-selection signal is robust enough to describe, but not promote.	CQR common-cell wins 58 of 94 cells; bootstrap selection count is cqr=1,000.	Report the fixed-GBM CQR pipeline as the largest selected-cell signal, with Mondrian and CV+ as secondary candidates under the current coverage gate.	CQR final-selection and universal-deployment readings require separate validation.
S1b	Check whether CQR's signal survives a backend-confound diagnostic.	Model-matched CQR completed 4,564 rows; paired cells=224; selected cells are fixed_gbm_cqr=116, model_matched_cqr=71, neither=37.	Use as a backend sensitivity check for the experiment-scoped CQR signal.	The check is backend-sensitivity evidence for the CQR pipeline signal.
S2	Check whether post-selection and bridge diagnostics preserve the same boundary.	Post-selection diagnostic counts are cqr=18 and mondrian_abs=7; the bridge slice has 45 completed atomic runs.	Use the slice as diagnostic stress-test evidence.	Do not upgrade the slice into study-wide method choice or validated Venn-Abers regression intervals.
S3	Check whether bounded-support endpoint behavior authorizes a validity claim.	0 bounded-support-validity-ready bundles and 11 raw endpoint-excursion bundles.	Report endpoint policy closure and retained domain caveats.	Bounded-support and endpoint-validity claims require separate validation.
S4	Check whether group diagnostics define a population-group-inference estimand.	187 pairwise group comparisons, 0 population-group-inference-ready bundles, and 0 population-estimand-declared bundles.	Use group rows as heterogeneity diagnostics.	Population-level and deployment group-inference claims require separate validation.
S5	Check whether duplicate and leakage controls strengthen or limit interpretation.	29 duplicate actions, 46 quarantined actions, 0 unquarantined actions, and hard leakage not detected in scanned artifacts.	Retain integrity caveats as part of the scientific record.	Integrity caveats remain part of the scientific record.
S6	Check how the evidence map supports source inspection.	3,643 KG nodes, 21,019 edges, 0 isolated nodes, and evidence scope recorded in the manifest.	Use the KG and package to trace claims to evidence.	Use the KG and site to inspect how claims connect to evidence.

3 S1. Method Selection Robustness Diagnostics

This section supports the main article by separating descriptive method behavior from study-wide method choice. In conformal prediction, `1 - alpha` is the target coverage level and `alpha` is the target miscoverage rate [3]. CQR builds an interval from lower and upper quantile models before calibration [7], while CV+ and jackknife-style methods use out-of-fold or leave-one-out predictions [1, 2].

Diagnostic item	Value	Scope
Candidate methods	3	CQR, CV+, and Mondrian absolute-residual calibration only
Common dataset-alpha cells	94	Shared comparison cells
Common-cell selected diagnostic method	`cqr`	Descriptive comparison label
Common-cell counts	cqr=58; cv_plus=15; mondrian_abs=21	Descriptive counts only
Bootstrap replicates	1,000	Selection stability diagnostic
Bootstrap counts	cqr=1,000	Selection-stability diagnostic only
Bootstrap primary selection rate	1.0000	Stability estimate, descriptive selection
Bootstrap primary selection-rate descriptive range	0.9962 to 1.0000	Precision check for the diagnostic rate
Leave-one-dataset retention	1.0000	Diagnostic robustness
Leave-one-alpha retention	1.0000	Diagnostic robustness
Final selection claim	`beyond this study`	Final selection remains beyond this study
Pairwise shared cells	94	Inferential support, not a claim upgrade
Main-result primary win rate	0.6778	Descriptive rate with final claim beyond this study
Main-result primary win-rate descriptive range	0.5757 to 0.7653	Uncertainty around the descriptive rate
Robustness common-cell primary win rate	0.6170	Common-cell diagnostic only
Robustness common-cell descriptive range	0.5160 to 0.7089	Precision check, not claim scope

Reader note: this section explains why the fixed-GBM CQR pipeline, Mondrian calibration, and CV+ can be reported as practical candidates within this experiment while the final-selection gate remains beyond this study. The descriptive ranges summarize uncertainty around the observed rates; they do not change the claim scope of the final-selection claim.

3.1 S1b. CQR Backend Sensitivity Check

The completed model-matched CQR rerun checks whether the historical fixed-GBM CQR backend explains the CQR signal. It preserves the experiment-scoped interpretation and does not create a method-selection claim.

Backend diagnostic item	Value	Scope
Fixed-GBM CQR completed rows	4,564	Historical comparator rows
Model-matched CQR completed rows	4,564	Completed rerun rows
Paired dataset-alpha-model-family cells	224	Direct CQR backend comparison cells
Fixed-GBM CQR selected cells	116	Coverage-eligible interval-score selections
Model-matched CQR selected cells	71	Coverage-eligible interval-score selections
Neither coverage-eligible variant	37	Cells where both CQR variants fail the coverage-eligibility rule
Study-wide method selection supported	`False`	No study-wide method selection follows from this table

4 S2. Post-Selection Validation Diagnostics

Post-selection validation asks whether the descriptive pattern persists in a held-out validation-style slice after the candidate methods have already been identified. The supplement treats this as a diagnostic stress test. It does not convert the observed counts into a study-wide method choice.

Diagnostic item	Value	Scope
Validation datasets	5	Narrow validation scope
Validation dataset-alpha cells	25	Shared cells only
Completed validation atomic runs	225	Completed diagnostic rows
Validation diagnostic counts	cqr=18; mondrian_abs=7	No study-wide method choice
Validation primary win rate	0.7200	Post-selection diagnostic rate
Validation primary win-rate descriptive range	0.5242 to 0.8572	Wide uncertainty remains visible
Feature leakage violations	0	Leakage sidecars clean in this slice
Width pathology rows	6	Retained caveat evidence
Bridge validation datasets	1	Dataset-final-gate bridge slice
Bridge validation atomic runs	45	Auxiliary diagnostic evidence
Bridge validation diagnostic counts	cqr=3; cv_plus=1; mondrian_abs=1	Bridge-slice diagnostics only
Venn-Abers bridge under-coverage runs	14	Negative bridge evidence

Reader note: post-selection diagnostics are stress tests of a descriptive pattern. They do not license language such as study-wide method choice, globally best conformal method, or validated

Venn-Abers regression interval. The Venn-Abers rows are especially narrow: they concern the evaluated bridge, while predictive-distribution and generalized Venn-Abers work remain separate literature objects [5, 4, 9, 6].

5 S3. Bounded-Support Endpoint Policy

Bounded-support checks ask whether prediction interval endpoints respect the natural domain of the target. This is different from ordinary marginal coverage. Weighted conformal ideas can address some covariate-shift settings [8], but the present endpoint policy is an empirical domain-validity gate. The current gate blocks bounded-support validity claims.

Diagnostic item	Value	Scope
Bounded-support bundles	15	Research Document-candidate bundles
Bounded-support datasets	6	Current endpoint-audit scope
Raw endpoint-excursion bundles	11	Domain blocker retained
Post-handling validated bundles	15	Policy triage complete
Clean or not-applicable bundles	4	Not enough for global validity
Validity-ready bundles	0	No bounded-support validity claim
stronger claim-ready bundles	0	Positive endpoint claim beyond this study
Endpoint support status counts	beyond this study_natural_domain_endpoint_excursions=11; clean_no_natural_domain_endpoint_excursions=3; not_applicable_unbounded_target_endpoint_hygiene_recorded=1	Support-status inventory, not a validity upgrade

Reader note: bounded-support evidence is stricter than ordinary coverage evidence. A method can have acceptable empirical coverage and still fail a natural-domain endpoint gate. The supplement therefore reports endpoint closure as a limitation, not as a post-processing success story.

6 S4. Group Diagnostics

Group coverage diagnostics compare interval behavior across observed groups. They are useful for auditing heterogeneity, but they are not population-level group inference claims unless the population estimand, protected-attribute scope, sampling design, and multiplicity treatment support that claim.

Diagnostic item	Value	Scope
Group diagnostic bundles	15	Diagnostic group scope
Group diagnostic datasets	6	Current diagnostic scope
Bootstrap replicates	500	Gap-uncertainty diagnostic
Group counts recorded	15	Required diagnostic metadata

Coverage by group recorded	15	Group comparison evidence
Width by group recorded	15	Interval-size comparison evidence
Gap uncertainty recorded	15	Multiplicity-aware caveat evidence
Pairwise group comparisons	187	Multiplicity scope, not group-inference proof
Population-estimand-declared bundles	0	Required for population-level group inference, absent here
Protected-attribute-scope-declared bundles	0	Required scope declaration, absent here
Weighted-estimand-applied bundles	0	No weighted population estimand applied
Sampling-weight-policy-declared bundles	15	Policy exists but does not open group-inference claim
Population-group-inference-ready bundles	0	No population-level group inference claim

Reader note: group rows can show where interval behavior differs, but group inference is an estimand-level claim. Because the current population estimand and protected-attribute scope are not declared at the bundle level, the correct supplement reading is diagnostic heterogeneity, not population-level group inference.

7 S5. Duplicate And Integrity Caveats

Duplicate and split-integrity checks protect the interpretation of model comparisons. If repeated rows, row signatures, or model-visible duplicates can affect train/calibration/test separation, the result must retain the caveat rather than become stronger prose.

Diagnostic item	Value	Scope
Duplicate actions	29	Caveat inventory scope
Open duplicate actions	0	Closure evidence only
Paired duplicate comparison rows	1,043	Sensitivity evidence
Quarantined actions	46	Stronger-claim validation retained
Unquarantined actions	0	No untracked duplicate action
Cross-run leakage status	`hard_leakage_not_detected_in_scanned_artifacts`	Hard leakage not detected in scanned artifacts
Unsupported claim hits	0	Claim scan clean for this audit
Cross-run risk counts	medium=39; pass=109	Risk inventory retained
Cross-run caveat counts	duplicate_cluster_plus_family_internal_fold_caveat=17; duplicate_signature_cross_split_caveat=14; model_visible_signature_cross_split_caveat=15	Caveats constrain interpretation

Reader note: the clean unsupported-claim scan does not erase duplicate or split caveats. It means the scanned artifacts did not contain unsupported claim language; the caveats still travel with the empirical interpretation.

8 S6. Traceability And Release State

The supplement is linked to a knowledge graph snapshot with 3,643 nodes, 21,019 edges, and 0 isolated nodes. The graph connects report sections, source artifacts, citations, interpretation scope, and quality checks. It is part of the Research Atlas evidence trail for inspecting the study.

Venn-Abers evidence is retained as part of the negative and boundary evidence for this study [5]. The supplement does not claim that the current interval bridge validates Venn-Abers regression intervals.

9 Interpretation Scope

- Method evidence is diagnostic; no study-wide method choice or deployment rule is established.
- Bounded-support endpoint results block validity claims.
- Group diagnostics do not establish population-level group inference.
- Duplicate and integrity caveats are retained rather than hidden.

10 References

- `@barber2020jackknife_plus`: https://arxiv.org/abs/1905.02928
- `@kim2020jackknife_after_bootstrap`: https://arxiv.org/abs/2002.09025
- `@lei2017distribution_free_regression`: https://arxiv.org/abs/1604.04173
- `@nouretdinov2018ivapd`: https://proceedings.mlr.press/v91/nouretdinov18a.html
- `@nouretdinov2024ivapd_applications`: https://proceedings.mlr.press/v230/nouretdinov24a.html
- `@petej2026inductive_venn_abers_regressors`: https://arxiv.org/html/2605.06646v1
- `@romano2019conformalized_quantile_regression`: https://arxiv.org/abs/1905.03222
- `@tibshirani2020covariate_shift`: https://arxiv.org/abs/1904.06019
- `@vanderlaan2025generalized_venn_abers`: https://proceedings.mlr.press/v267/van-der-laan25a.html

11 Source Artifacts

- `main_article_draft`: `study/research_document/main_article_draft.json`
- `individual_experiment_report_draft`: `study/research_document/individual_experiment_report_draft.json`
- `article_supplement_blueprint_alignment`: `study/research_document/article_supplement_blueprint_alignment.json`
- `method_selection_robustness_audit`: `study/reports/methodology_sanity_audit_20260627/method_selection_robustness_audit.json`

- method_selection_inferential_audit: study/reports/methodology_sanity_audit_20260627/method_selection_inferential_audit.json
- method_selection_post_selection_validation_results: study/reports/methodology_sanity_audit_20260627/method_selection_post_selection_validation_results.json
- dataset_final_gate_post_selection_validation_bridge_results: study/reports/methodology_sanity_audit_20260627/dataset_final_gate_post_selection_validation_bridge_results.json
- bounded_support_endpoint_closure_audit: study/reports/methodology_sanity_audit_20260627/bounded_support_endpoint_closure_audit.json
- bounded_support_positive_validation_protocol: study/research_document/bounded_support_positive_validation_protocol.json
- group inference_group_diagnostic_audit: study/reports/methodology_sanity_audit_20260627/group inference_group_diagnostic_audit.json
- group inference_population_readiness_audit: study/reports/methodology_sanity_audit_20260627/group inference_population_readiness_audit.json
- group inference_group_multiplicity_scope: study/research_document/group inference_group_multiplicity_scope.json
- duplicate_sensitivity_closure_audit: study/reports/methodology_sanity_audit_20260627/duplicate_sensitivity_closure_audit.json
- duplicate_content_quarantine_audit: study/reports/methodology_sanity_audit_20260627/duplicate_content_quarantine_audit.json
- cross_run_integrity_audit: study/reports/methodology_sanity_audit_20260627/cross_run_integrity_audit.json
- publication_citation_registry: study/research_document/publication_citation_registry.json
- knowledge_graph_quality_summary: study/reports/knowledge_graph_quality/quality_summary.json
- cqr_fixed_vs_model_matched_synthesis: study/reports/model_matched_cqr_rerun_plan/cqr_fixed_vs_model_matched_synthesis.json

References

[1] Rina Foygel Barber, Emmanuel J. Candes, Aaditya Ramdas, and Ryan J. Tibshirani. Predictive inference with the jackknife+, 2020.

[2] Byol Kim, Chen Xu, and Rina Foygel Barber. Predictive inference is free with the jackknife+-after-bootstrap, 2020.

[3] Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression, 2017.

[4] Ilia Nouretdinov and James Gammerman. Inductive venn-abers predictive distributions: New applications & evaluation. In *Proceedings of the Thirteenth Symposium on Conformal and Probabilistic Prediction with Applications*, volume 230 of *Proceedings of Machine Learning Research*, pages 490–507. PMLR, 2024.

[5] Ilia Nouretdinov, Denis Volkhonskiy, Pitt Lim, Paolo Toccaceli, and Alexander Gammerman. Inductive venn-abers predictive distribution. In *Proceedings of the Seventh Workshop on Conformal and Probabilistic Prediction and Applications*, volume 91 of *Proceedings of Machine Learning Research*, pages 15–36. PMLR, 2018.

[6] Ivan Petej and Vladimir Vovk. Inductive venn-abers and related regressors, 2026.

[7] Yaniv Romano, Evan Patterson, and Emmanuel J. Candes. Conformalized quantile regression, 2019.

[8] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candes, and Aaditya Ramdas. Conformal prediction under covariate shift, 2020.

[9] Lars Van Der Laan and Ahmed Alaa. Generalized venn and venn-abers calibration with applications in conformal prediction. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 60748–60763. PMLR, 2025.